

SkillBridge AI

Concise Project Reference

Flutter | Firebase | ML Engine | Gemini AI | SDG 8

SkillBridge AI is not just a job-finding app.

It is a **job app + skill-gap checker + learning guide + career coach** in one place. It helps users see **which job fits them, what skills they are missing, and what to learn next.** That is the main special thing.

Feature	SkillBridge AI	Shomvob Jobs	Bdjobs	LinkedIn	Typical job apps
Find jobs	Yes	Yes	Yes	Yes	Yes
Job matching	Advanced AI matching	Basic/AI matching	Search/filter based	Profile + AI/search based	Usually basic
Skill-gap analysis	Yes	No clear full gap analysis shown	No	Limited	Rare
Tells what skills are missing	Yes	No	No	Limited	Rare
Suggests courses for missing skills	Yes	Training available	Career support, not full skill-gap based	Learning available	Rare
Personalized learning path	Yes	Partial	Partial	Partial	Rare
Career transition planner	Yes	No clear feature	No	Limited	Rare
Career confidence tracker	Yes	No	No	No	No
MCQ skill assessment	Yes	No	No clear feature	No	Rare
CV upload and skill extraction	Yes	Not clearly shown	Common resume/apply tools	Yes	Sometimes
AI career chatbot	Yes	No clear feature	No clear feature	Some AI features	Rare
Bangladesh-focused job insights	Yes	Yes	Yes	Not mainly	Sometimes
Geographic job insights in Bangladesh	Yes	No clear feature	No clear feature	No	Rare
Workforce/market skill trends	Yes	No clear feature	No clear feature	Some market insight	Rare
Job alerts	Yes	Likely yes	Yes	Yes	Often yes
Networking	Limited	Some professional network focus	Low	Very strong	Usually low

In very simple words

- **SkillBridge AI**: best for someone who wants **guidance**, not only job posts.
- **Shomvob Jobs**: good for **Bangladesh job search + training + hiring platform**.
- **Bdjobs**: good for **large number of Bangladesh job listings**, alerts, saved filters, and career services.
- **LinkedIn**: best for **professional networking**, profile building, and global jobs; it also has learning and some AI job-search tools.

1. Project Overview

SkillBridge AI is a Flutter mobile app that helps Bangladeshi students and job seekers find the right jobs, understand their skill gaps, and plan how to improve. It is not just a job board — it is a full career guidance system backed by 8 research papers and aligned with UN SDG 8 (Decent Work and Economic Growth).

Item	Detail
App Name	SkillBridge AI
Platform	Flutter — Android & iOS
Auth & Database	Firebase Auth + Firestore
AI Chatbot	Google Gemini 2.5 Flash
ML Engine	Pure Dart — runs on-device, works offline
State Management	Provider / ChangeNotifier
Design	Material 3, Plus Jakarta Sans font, Blue #1A56DB
Target Users	Bangladeshi students and career changers
Dataset	1,000 student records + 50,000 job postings + 10,000 job listings
Research Papers	8 peer-reviewed papers

2. Folder Structure

All code lives inside the lib/ folder, split into 7 sub-folders with 49 Dart files total.

Folder	What it contains
lib/	App entry point (main.dart) and Firebase config
lib/screens/	All 25 UI screens the user sees
lib/screens/auth/	4 login/register/verification screens
lib/ml/	8 machine learning algorithm files
lib/data/	6 files with job data, courses, and research models
lib/models/	1 file — the core CareerProfile user model
lib/services/	3 files — AppState, Gemini API, SBERT service
lib/theme/	3 files — colours, fonts, shared widgets

3. All Screen Files — Simple Explanations

Auth Screens (lib/screens/auth/)

File	What it does
login_screen.dart	Login page. Supports email/password, Google, LinkedIn, and fingerprint/face login.
register_screen.dart	Sign-up page. Collects name, email, password, field of study, and initial skills. Saves photo. Creates Firebase account.
email_verification_screen.dart	Shown after sign-up. Waits for the user to click the verification email. Auto-detects when done.
forget_password_screen.dart	Enter your email, get a password reset link. Simple animated screen.

Navigation & Home

File	What it does
main.dart	App start point. Sets up Firebase, locks portrait mode, loads saved data, defines all screen routes.
main_nav.dart	Main shell with 5 bottom tabs (Home, Jobs, Learning, Dashboard, Chatbot) and a side drawer with all other screens.
home.dart	Home tab. Shows greeting, quick-action cards (Find Jobs, CV Upload, Courses, Chatbot, Insights).

Core Job & Skills Screens

File	What it does
job_result.dart	Shows AI-matched jobs with a colour-coded match score. Tap a job to see your skill gap for it.
skill_gap.dart	Deep analysis per job. Shows skills you have (green), skills missing (red), readiness %, and course suggestions.
skill_input.dart	Manually add/remove your skills. High-value skills get a 2x weight badge. Saves and runs ML scoring.
job_transition_screen.dart	Pick a current job and a target job. Shows overlap %, transition difficulty, and which skills to learn first.
job_alerts_screen.dart	Create saved job alerts with keyword/location/type filters and notification frequency (instant/daily/weekly).

Learning & Courses Screens

File	What it does
browse_courses_screen.dart	Browse the full course catalogue. Filter by category, format (video/book), and cost (free/paid). Opens course URLs.

learning.dart	Personalised course tab. Ranks courses based on your learning style preferences (short vs long, theory vs examples).
career_guide.dart	Shows recommended career paths for your field of study, industry match scores, career tips, and success estimates.

Analytics & Insights Screens

File	What it does
dashboard_screen.dart	Central dashboard. Readiness score ring, skill match chart, application summary, and recommended jobs.
assessment_hub_screen.dart	MCQ quizzes by topic and difficulty. Shows score ring and subtopic breakdown after each quiz.
confidence_tracker_screen.dart	Self-rate your career confidence by category. Shows line charts of how your confidence changes over time.
skill_trends_screen.dart	Shows rising skills, year-on-year demand growth, and AI adoption rates by industry.
workforce_insights_screen.dart	Multi-generational workforce profiles: Gen Z, Millennial, Gen X, Boomer — skills, roles, and bridge skills.
geo_insights_screen.dart	City-wise job demand map for Bangladesh (Dhaka, Chittagong, Sylhet, etc.) with salary data in BDT.
application_tracker_screen.dart	Kanban board to track job applications: Wishlist, Applied, Interview, Offer, Rejected. Saved locally.

Profile, CV & Settings Screens

File	What it does
profile_input.dart	Edit name, field of study, experience, industry, salary expectation, location, bio, and photo. Shows completeness bar.
cv_upload_screen.dart	Upload your CV (PDF/DOCX/TXT). App auto-extracts your skills and runs job matching. 5-step animated pipeline.
chatbot_screen.dart	AI career chatbot powered by Gemini 2.5 Flash. You can attach your CV and ask it career questions.
privacy_settings_screen.dart	View all data the app collects, toggle consent settings, and see an activity audit log.

4. Machine Learning Files (lib/ml/)

All ML runs on your device in pure Dart — no internet needed. An optional Python server handles advanced SBERT re-ranking.

File	What it does
recommender.dart	The main ML engine. Runs 10-step pipeline: TF-IDF + Cosine + Jaccard + Skill Coverage + BM25 + Temporal Decay + MMR diversity + Cross-

	domain transfer + Bias audit + Percentile ranking. Outputs match scores with explanations.
tfidf.dart	Converts skills and job descriptions into number vectors. Handles synonyms (js=javascript), skill boosting (2x weight for known skills), stopword removal, and BM25 frequency dampening.
cosine.dart	Measures how similar two vectors are. Score 0-1. Uses L2 normalisation and attention weighting. Tiers: Strong (≥ 0.75), Moderate (0.40-0.74), Weak (< 0.40).
jaccard.dart	Measures skill set overlap. How many of your skills appear in the job requirements. Used alongside cosine as a second signal.
sbert.dart	Offline SBERT approximation. 32-dim cluster embeddings built from Bangladesh job market vocabulary. Works with zero internet, zero latency.
rca_calculator.dart	Revealed Comparative Advantage. Measures how much more important a skill is for a specific job vs the overall market. Tiers: Critical, Core, Supporting, Peripheral.
skills_space.dart	Measures similarity between two job roles using skill co-occurrence patterns. Powers the transition difficulty score (Easy/Medium/Hard).
skill_similarity.dart	Computes how often two skills appear together in the same job. Cached using symmetric pair-key hashing for speed.
skill_prioritizer.dart	Ranks missing skills by priority = Importance x Distance. Labels: Already Have, Learn First, Learn Soon, Optional.

5. Data Files (lib/data/)

File	What it contains
jobs.dart	All job listings. Typed enums for JobType (full-time, part-time, contract, intern) and JobLevel (entry to lead). Data from 50,000 job postings across 7 industries.
courses.dart	Full course catalogue. Typed enums for length, detail level, format (video/book/slide/webpage), and learning strategy. Each course has provider, URL, skills, cost, and rating.
career_guidance_data.dart	Analysis of 1,000 student records. Career path recommendations per field of study. Industry match scores, career tips, and success probabilities.
fairness_monitor.dart	Bias detection engine. Flags masculine-coded words, feminine-coded words, and protected-attribute proxies in job descriptions. Uses Shannon entropy to check if recommendations are too concentrated in one industry.
intention_tracker.dart	Tracks your employment intention over time (Study Abroad / Further Education / Seek Employment / Start Business / Undecided). Shows how your intention changes month by month.
learner_preferences.dart	15-dimensional preference vector (from Tavakoli 2022 research). Stores what kind of learning content you prefer. Used to rank courses personally for you.

6. Models (lib/models/)

career_profile.dart

The main user data model. Stores name, email, field of study, skills, experience level, target industry, expected salary, location, bio, and profile photo path. Contains typed enums for FieldOfStudy (11 options), ExperienceLevel (student to lead), and TargetIndustry (11 industries). Supports JSON save/load and immutable copyWith updates.

7. Services (lib/services/)

File	What it does
app_state.dart	Central brain of the app. Holds all user data — profile, skills, preferences, confidence scores, intention history, theme mode, saved jobs, and application tracker entries. Saves everything to device storage. Runs the ML readiness score on every skill change.
gemini_chat_service.dart	Handles all communication with Gemini 2.5 Flash API. Sends full chat history each turn. Limits context to ~175,000 tokens. Retries up to 3 times on failure. Has cancel token to stop mid-request. Detects rate limits and safety blocks.
sbert_service.dart	Sends top-50 TF-IDF results to a Python Flask server for SBERT re-ranking. Hybrid score = 40% TF-IDF + 45% SBERT + 15% Jaccard. Falls back to offline Dart SBERT if server is unreachable. App never crashes if server is down.

8. Theme Files (lib/theme/)

File	What it does
app_theme.dart	Global design system. Defines blue colour palette, Plus Jakarta Sans typography, dark and light Material 3 themes, button styles, card styles, and input styles.
app_widgets.dart	Shared reusable widgets used across many screens — score badges, skill chips, match rings (CustomPainter), shimmer loaders, and empty state illustrations.
confidence_tracker_screen_fixed.dart	An alternate fixed version of the confidence tracker screen kept in theme/ during iterative UI development.

9. Functional Requirements — What the App Must Do

These are the 59 specific features the app must have.

Authentication (FR-01 to FR-08)

ID	Requirement
FR-01	Register with email, password, name, field of study
FR-02	Verify email before entering the app
FR-03	Login with email and password
FR-04	Login with Google
FR-05	Login with LinkedIn
FR-06	Login with device biometrics (fingerprint/face)
FR-07	Reset password via email
FR-08	Logout with confirmation dialog

Profile & CV (FR-09 to FR-17)

ID	Requirement
FR-09	Create and edit career profile
FR-10	Upload profile photo from gallery
FR-11	Set field of study, experience level, target industry, location
FR-12	Show profile completeness bar and achievement badges
FR-13	Profile data must persist across sessions
FR-14	Upload CV in PDF, DOCX, DOC, or TXT format
FR-15	Extract text from the uploaded CV
FR-16	Auto-populate extracted skills into profile
FR-17	Show 5-step animated analysis during CV processing

Job Matching & Skill Gap (FR-18 to FR-27)

ID	Requirement
FR-18	Recommend jobs using TF-IDF + Cosine Similarity
FR-19	Show match score per job (percentage + colour badge)
FR-20	Use Jaccard overlap score alongside cosine
FR-21	SBERT re-ranking must improve top results semantically
FR-22	Cover at least 7 industry sectors in recommendations
FR-23	Navigate from job card to its skill gap analysis

FR-24	Show which skills user has and which are missing per job
FR-25	Compute and show asymmetric skill coverage score
FR-26	Show course suggestions for each missing skill
FR-27	Show overall readiness % with colour-coded tips

Learning & Courses (FR-28 to FR-35)

ID	Requirement
FR-28	Browse full course catalogue
FR-29	Filter by category, format, and cost
FR-30	Real-time search with debounce
FR-31	Rank courses by 15-dim learner preference vector
FR-32	Enrol button opens external course URL
FR-33	Recommend career paths by field of study
FR-34	Show industry match scores via cosine similarity
FR-35	Show career tips and success probability estimates

Transition, Assessments & Tracking (FR-36 to FR-53)

ID	Requirement
FR-36	Select current and target job for transition planning
FR-37	Show skill overlap % and transition difficulty
FR-38	Rank missing skills by acquisition priority
FR-39	Show course recommendations per missing skill
FR-40	Take MCQ quizzes by topic and difficulty level
FR-41	Show score ring and subtopic breakdown after quiz
FR-42	Self-rate confidence by category
FR-43	Show confidence trends on line chart over time
FR-44	Dashboard: readiness score, pipeline summary, recommended jobs
FR-45	Show rising skills with YoY change data
FR-46	Show city-wise job demand for Bangladesh
FR-47	Show multi-generational workforce skill profiles
FR-48	Create job application cards
FR-49	Move cards through Kanban stages
FR-50	Application data persists across sessions
FR-51	Create job alerts with keyword/location/type filters

FR-52	Set alert frequency: Instant, Daily, or Weekly
FR-53	Alerts show match count and support sorting

Chatbot & Privacy (FR-54 to FR-59)

ID	Requirement
FR-54	Multi-turn career chatbot powered by Gemini
FR-55	Attach CV files to chatbot conversation
FR-56	Chatbot remembers conversation history within session
FR-57	Show all data categories collected by the app
FR-58	Toggle analytics consent and personalisation
FR-59	Show activity audit log

10. Non-Functional Requirements — How Well It Must Work

These describe quality, speed, security, and design standards — not features, but how the features must behave.

Performance (NFR-01 to NFR-05)

ID	Requirement
NFR-01	TF-IDF + Cosine job matching must finish in under 500ms on-device
NFR-02	SBERT API call must timeout in 8 seconds maximum
NFR-03	App must never freeze during ML computation — use async
NFR-04	CV text extraction must not block the UI thread
NFR-05	Course search debounce must be 300ms

Scalability (NFR-06 to NFR-08)

ID	Requirement
NFR-06	TF-IDF engine must support adding documents incrementally
NFR-07	Recommendation engine must handle 50,000+ job listings
NFR-08	Skill similarity cache must use symmetric pair-key hashing to avoid repeated computation

Reliability (NFR-09 to NFR-12)

ID	Requirement
NFR-09	App must never crash if SBERT server is down — fall back to offline
NFR-10	Gemini API failures must show clear error messages

NFR-11	Firebase auth failures must show specific user-friendly messages
NFR-12	All saved data reads must handle missing or corrupt data gracefully

Security (NFR-13 to NFR-16)

ID	Requirement
NFR-13	Gemini API key must be injected via --dart-define, never hardcoded
NFR-14	Firebase config must not be in public version control for production
NFR-15	Authentication must use Firebase Auth with email verification
NFR-16	Biometric auth must use the device secure enclave via local_auth

Usability & Design (NFR-17 to NFR-22)

ID	Requirement
NFR-17	Full dark mode and light mode, toggled globally
NFR-18	All screens must respect the global ThemeMode from AppState
NFR-19	Plus Jakarta Sans typography throughout the entire app
NFR-20	All match scores colour-coded: green (strong), orange (moderate), red (weak)
NFR-21	Animations via flutter_animate for consistent timing
NFR-22	App locked to portrait orientation only

Code Quality (NFR-23 to NFR-26)

ID	Requirement
NFR-23	Every ML algorithm must cite its research paper in code comments
NFR-24	All categorical data must use typed enums, not plain strings
NFR-25	Value objects must be immutable using @immutable annotation
NFR-26	Each file must have one clear responsibility only

Ethics & Compliance (NFR-27 to NFR-33)

ID	Requirement
NFR-27	Detect gender-coded language in job descriptions (Gaucher et al. 2011)
NFR-28	Enforce industry diversity using Shannon entropy scoring
NFR-29	Exclude protected-attribute proxies from skill matching
NFR-30	App must explicitly align with UN SDG 8
NFR-31	Minimum touch target sizes on all interactive elements

NFR-32	Sufficient colour contrast in both dark and light modes
NFR-33	All currency values displayed in BDT (Bangladeshi Taka)

11. ML Algorithms — Quick Reference

Algorithm	What it does	Where used
TF-IDF	Turns skills/jobs into number vectors	tfidf.dart
Cosine Similarity	Measures vector closeness (0-1 score)	cosine.dart, recommender.dart
Jaccard Coefficient	Measures skill set overlap	jaccard.dart
Asymmetric Coverage	How much of the job skills does user cover	jaccard.dart, skill_gap.dart
BM25	Stops common words from dominating the score	tfidf.dart
MMR Diversity	Ensures results span multiple industries	recommender.dart
RCA	How critical a skill is for a specific job	rca_calculator.dart
SKILLS SPACE Theta	Occupation-level similarity score	skills_space.dart
Skill Co-occurrence	How often two skills appear together	skill_similarity.dart
Skill Prioritiser	Ranks missing skills: Learn First / Learn Soon	skill_prioritizer.dart
Local SBERT	Offline 32-dim semantic embedding	sbert.dart
SBERT Hybrid	0.40 TF-IDF + 0.45 SBERT + 0.15 Jaccard	sbert_service.dart
Dot-product Prefs	Ranks courses by learning style match	learner_preferences.dart
Shannon Entropy	Checks if recommendations are too one-sided	fairness_monitor.dart
Temporal Decay	Boosts newer job postings slightly	recommender.dart

12. Research Papers (8 Papers)

Paper	What it contributes
Ajjam & Al-Raweshidy (2026)	TF-IDF + cosine job matching. Greedy one-to-one matching. Cross-domain transfer. Bias audit. Core algorithm basis.
Alsaif et al. (2022)	Skill weighting (2x for skills). Jaccard vs cosine comparison. Tokenisation pipeline.
Tavakoli et al. (2022)	15-dim learner preference vector. Dot-product course scoring. Content taxonomy (length/detail/format).
Dawson et al. (2021)	RCA skill importance. Occupation similarity Theta. Job transition probability.
Zhisheng Chen (2022)	AI-human recruitment model. 6-stage hiring pipeline. Person-Job Fit. Bias removal.
Li Huang (2022)	Employment intention classification. 5 categories. Longitudinal intention tracking.

Xiao & Zheng (2025)	Career confidence self-assessment. Category-based tracking over time.
Alaql et al. (2023)	Multi-generational workforce analysis. Gen Z, Millennial, Gen X, Boomer profiles.

13. Technology Stack

Layer	Technology
Mobile Framework	Flutter (Dart)
State Management	Provider + ChangeNotifier
Auth & Database	Firebase Auth + Firestore
AI Chatbot	Google Gemini 2.5 Flash (REST API)
On-device ML	Pure Dart — TF-IDF, Cosine, Jaccard, RCA, SBERT
SBERT Server	Python Flask microservice
Charts	fl_chart
Animations	flutter_animate
Fonts	Plus Jakarta Sans + Google Fonts
File Handling	file_picker, syncfusion_flutter_pdf, archive
Social Auth	google_sign_in, flutter_web_auth_2, local_auth
Local Storage	SharedPreferences
HTTP	package:http
Firebase Project	skill-bridge-ai-c02ef

14. Dependencies — pubspec.yaml

The following third-party packages are used in SkillBridge AI. All are managed via Flutter's pub.dev package system.

Package	Version	Purpose in SkillBridge AI
provider	^6.1.1	Global state management — AppState ChangeNotifier shared across all screens.
shared_preferences	^2.3.0	Persists user data (profile, skills, preferences, theme) between app restarts.
google_sign_in	^6.2.1	Google OAuth 2.0 login on the Login screen.

local_auth	^2.3.0	Fingerprint and Face ID biometric authentication on the Login screen.
flutter_web_auth_2	^4.0.0	LinkedIn OAuth via web redirect flow on the Login screen.
fl_chart	^1.1.1	All charts in the app: line charts (readiness history), bar charts (industry coverage), radar charts (confidence), pie charts (workforce insights).
flutter_animate	^4.3.0	Smooth staggered card animations, fade-in, slide-up effects across all screens.
lottie	^3.0.0	JSON-based Lottie animation files for loading states and illustrations.
shimmer	^3.0.0	Skeleton shimmer loading effect on cards while results are being computed.
word_cloud	1.0.3	Skills word cloud visualization on the Skill Trends screen.
google_fonts	^8.0.2	Downloads Plus Jakarta Sans and Poppins fonts from Google Fonts CDN.
http	^1.2.0	HTTP client for Gemini API calls in gemini_chat_service.dart.
cached_network_image	^3.3.1	Caches remote images (course thumbnails, provider logos) for fast loading.
intl	^0.20.2	Formats BDT currency values and dates in Bangladeshi locale throughout the app.
file_picker	^10.3.10	Opens the device file chooser for CV upload on CV Upload and Chatbot screens.
syncfusion_flutter_pdf	^33.1.46	Extracts text from PDF resumes on-device. PdfDocument + PdfTextExtractor.
archive	^3.6.1	Unzips DOCX files (which are ZIP archives) to read document.xml for text extraction.
url_launcher	^6.3.0	Opens course enrollment URLs and external links in the device browser.
image_picker	^1.1.2	Allows user to pick a profile photo from the device gallery.